

Example 2: A Biography (with planted contradictions)

FactReasoner — Ideation

June 16, 2026

1 A biography

Source paragraph

Lanny Flaherty is an American actor born on December 18, 1949, in Pensacola, Florida. He has appeared in numerous films, television shows, and theater productions throughout his career, which began in the late 1970s. Some of his notable film credits include “King of New York,” “The Abyss,” “Natural Born Killers,” “The Game,” and “The Straight Story.” On television, he has appeared in shows such as “Law & Order,” “The Sopranos,” “Boardwalk Empire,” and “The Leftovers.” Flaherty has also worked extensively in theater, including productions at the Public Theater and the New York Shakespeare Festival. He is known for his distinctive looks and deep gravelly voice, which have made him a memorable character actor in the industry.

Atomic units (in order of appearance)

- F1.** Lanny Flaherty is an American actor.
- F2.** Lanny Flaherty was born on December 18, 1949.
- F3.** Lanny Flaherty was born in Pensacola, Florida.
- F4.** Lanny Flaherty has appeared in numerous films, television shows, and theater productions.
- F5.** Lanny Flaherty’s career began in the late 1970s.
- F6.** Lanny Flaherty’s notable film credits include “King of New York.”
- F7.** Lanny Flaherty’s notable film credits include “The Abyss.”
- F8.** Lanny Flaherty’s notable film credits include “Natural Born Killers.”
- F9.** Lanny Flaherty’s notable film credits include “The Game.”
- F10.** Lanny Flaherty’s notable film credits include “The Straight Story.”
- F11.** Lanny Flaherty has appeared in the television show “Law & Order.”
- F12.** Lanny Flaherty has appeared in the television show “The Sopranos.”
- F13.** Lanny Flaherty has appeared in the television show “Boardwalk Empire.”

- F14.** Lanny Flaherty has appeared in the television show “The Leftovers.”
- F15.** Lanny Flaherty has worked extensively in theater.
- F16.** Lanny Flaherty’s theater work includes productions at the Public Theater.
- F17.** Lanny Flaherty’s theater work includes productions at the New York Shakespeare Festival.
- F18.** Lanny Flaherty is known for his distinctive looks and deep gravelly voice.
- F19.** Lanny Flaherty’s distinctive looks and voice have made him a memorable character actor.

Relations

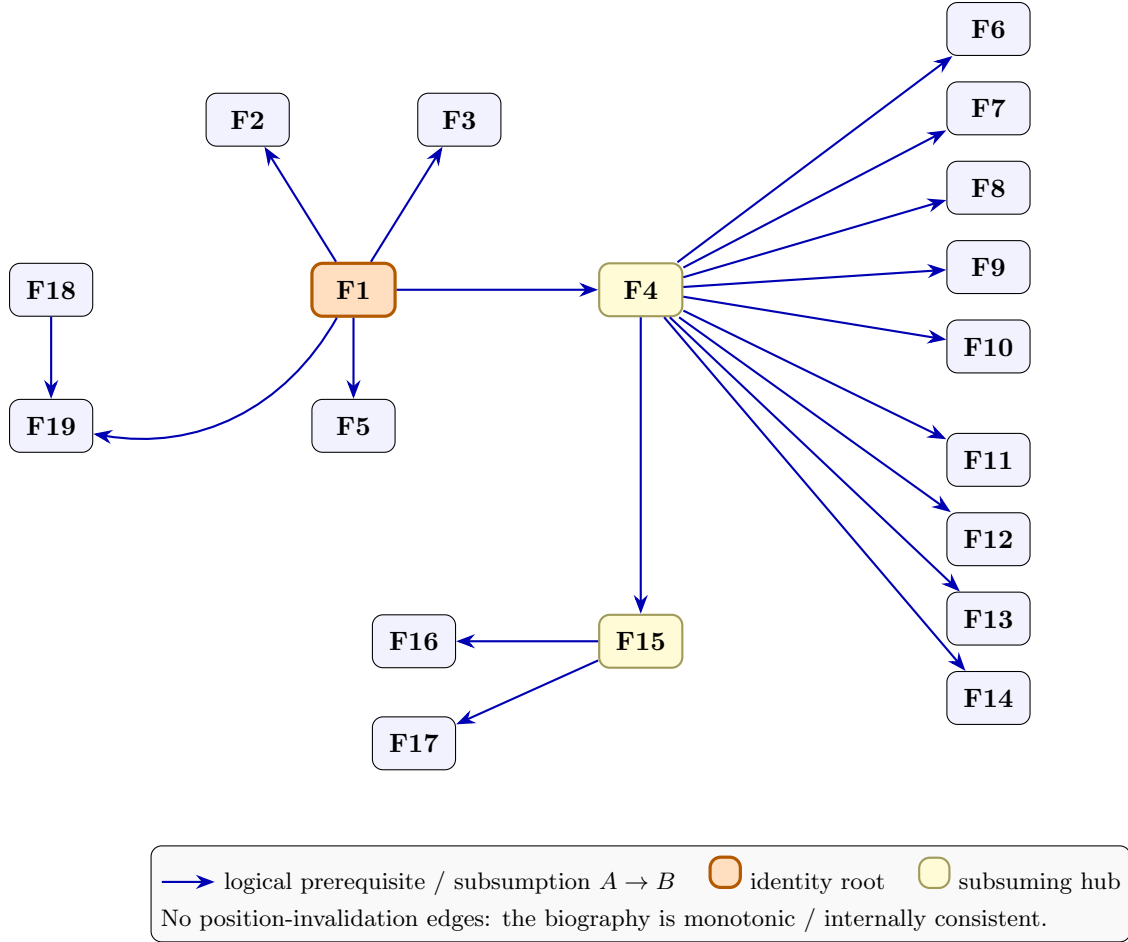
Logical prerequisites ($A \rightarrow B$). Here the prerequisite relation is mostly one of *subsumption* / *instantiation*: the general claim that he appeared in films, TV, and theater (F4) underpins each specific credit, and the identity claim (F1) underpins everything.

- $F1 \rightarrow F4$ — being an actor is presupposed by having a body of acting work.
- $F1 \rightarrow F5$ — being an actor is presupposed by having an acting career.
- $F4 \rightarrow F6, F7, F8, F9, F10$ — the general “appeared in films” claim subsumes each specific film credit.
- $F4 \rightarrow F11, F12, F13, F14$ — the general “appeared in TV shows” claim subsumes each specific show.
- $F4 \rightarrow F15$ — the general claim subsumes the theater-work claim.
- $F15 \rightarrow F16, F17$ — working in theater is presupposed by the specific theater venues.
- $F18 \rightarrow F19$ — the distinctive looks/voice are the stated cause of his being a memorable character actor.
- $F1 \rightarrow F19$ — being an actor is presupposed by being a memorable *character* actor.

Position invalidations ($A \neq B$). *None.* Unlike the legal example, this biography is monotonic: every later claim is consistent with the earlier ones, and no statement retracts or contradicts a prior position. The absence of invalidation edges is itself the finding.

Relation graph

All edges are logical prerequisites / subsumption ($A \rightarrow B$); there are no invalidation edges. The identity node **F1** is the root; the “body of work” node **F4** fans out to the specific credits.



A modified biography with planted contradictions

The biography above is internally consistent, so it has no invalidation edges. To illustrate the contradiction relation we now *alter* the passage, inserting later statements that conflict with earlier ones (different birth year and place, a different nationality, a different career-start decade, and a voice description that reverses the earlier one). Each contradiction is a positional invalidation: the earlier atom, asserted first, is contradicted by a later atom.

Modified source paragraph (contradictions in *italics*).

Lanny Flaherty is an American actor born on December 18, 1949, in Pensacola, Florida. His career began in the late 1970s, and he appeared in numerous films, television shows, and theater productions. He is known for his deep gravelly voice. *Born in 1942, Flaherty was a Mississippi native who first appeared on screen only in the 1990s.* Critics often described *the British character actor* for his soft, high-pitched voice, which made him instantly recognizable.

Atomic units (self-contained, in order of appearance).

M1. Lanny Flaherty is an American actor.

M2. Lanny Flaherty was born on December 18, 1949.

- M3.** Lanny Flaherty was born in Pensacola, Florida.
- M4.** Lanny Flaherty's career began in the late 1970s.
- M5.** Lanny Flaherty appeared in numerous films, television shows, and theater productions.
- M6.** Lanny Flaherty is known for his deep gravelly voice.
- M7.** Lanny Flaherty was born in 1942.
- M8.** Lanny Flaherty was a Mississippi native.
- M9.** Lanny Flaherty first appeared on screen only in the 1990s.
- M10.** Lanny Flaherty is a British character actor.
- M11.** Lanny Flaherty has a soft, high-pitched voice.
- M12.** Lanny Flaherty's voice made him instantly recognizable.

Relations

Logical prerequisites ($A \rightarrow B$).

- $M1 \rightarrow M5$ — being an actor is presupposed by having a body of acting work.
- $M1 \rightarrow M4$ — being an actor is presupposed by having an acting career.
- $M11 \rightarrow M12$ — having a (distinctive) voice is presupposed by that voice making him recognizable.

Position invalidations ($A \neq B$). Each later atom contradicts an earlier one asserted at a different position.

- $M2 \neq M7$ — born December 18, 1949 contradicts born in 1942.
- $M3 \neq M8$ — born in Pensacola, Florida contradicts being a Mississippi native.
- $M4 \neq M9$ — a career beginning in the late 1970s contradicts first appearing on screen only in the 1990s.
- $M1 \neq M10$ — being an *American* actor contradicts being a *British* character actor.
- $M6 \neq M11$ — a deep gravelly voice contradicts a soft, high-pitched voice.

Note. The contradictions are pairwise and local: each invalidation links exactly one early biographical claim to its later mutually exclusive restatement. The prerequisite structure (identity $\rightarrow$ work / career, voice $\rightarrow$ recognizability) is unaffected; M5 has no specific-credit children here because the modified paragraph drops the individual titles.

Relation graph. Solid blue arrows are logical prerequisites ($A \rightarrow B$). Dashed red arrows are position invalidations ($A \neq B$): each links an earlier (*green, asserted-first*) atom to the later (*red*) atom that contradicts it. The five contradiction pairs are stacked so no arrow crosses a node.

